

Problem 2c: Invalid Solution (Why the Wrong Guess Fails)

ECO 310 - Learning from Mistakes

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1 What Problem 2c Asks

Question: Show that you get an invalid solution to (b) if $p_1 < p_2$. And explain why intuitively, the guess was wrong about which nonnegativity constraint binds.

Translation: Use the SAME guess from part (b), but now assume $p_1 < p_2$ (good 1 is cheaper). Show that this leads to a contradiction.

2 Recall: The Guess from Part 2b

The guess was:

- Utility constraint is binding: $x_1 + x_2 = u^*$
- Nonnegativity on good 1 is binding: $x_1 = 0$
- Nonnegativity on good 2 is non-binding: $x_2 > 0$

This means: buy only good 2, so $x_1 = 0$ and $x_2 = u^*$.

3 Apply the Same Logic with $p_1 < p_2$

Let's follow the exact same steps as part (b), but now assume $p_1 < p_2$ (good 1 is CHEAPER).

3.1 Step 1: Use Complementary Slackness

Since $x_2 > 0$:

$$\lambda_2 = 0 \tag{1}$$

3.2 Step 2: Solve for λ_3

From the FOC for x_2 (which must equal zero since $x_2 > 0$):

$$-p_2 + \lambda_2 + \lambda_3 = 0 \tag{2}$$

$$-p_2 + 0 + \lambda_3 = 0 \tag{3}$$

$$\lambda_3 = p_2 \tag{4}$$

3.3 Step 3: Solve for λ_1

From the FOC for x_1 :

$$-p_1 + \lambda_1 + \lambda_3 \leq 0 \quad (5)$$

Substitute $\lambda_3 = p_2$:

$$-p_1 + \lambda_1 + p_2 \leq 0 \quad (6)$$

$$\lambda_1 \leq p_1 - p_2 \quad (7)$$

At the optimum, assume this binds:

$$\lambda_1 = p_1 - p_2 \quad (8)$$

3.4 Step 4: Check Validity

Now here's the problem! We assumed $p_1 < p_2$, so:

$$\lambda_1 = p_1 - p_2 \quad (9)$$

$$< 0 \quad (\text{since } p_1 < p_2) \quad (10)$$

INVALID!

Problem: We got $\lambda_1 < 0$, which violates the KKT condition that all Lagrange multipliers must be non-negative!

$$\lambda_1 \geq 0 \text{ is required, but we got } \lambda_1 = p_1 - p_2 < 0$$

This means the guess $x_1 = 0$, $x_2 = u^*$ is NOT optimal when $p_1 < p_2$.

4 Why Is This Invalid?

The Rule

KKT Condition: All Lagrange multipliers must satisfy $\lambda_i \geq 0$.
If any $\lambda_i < 0$, the solution is not optimal.

In our case:

- We found $\lambda_1 = p_1 - p_2$
- When $p_1 < p_2$, this gives $\lambda_1 < 0$
- This violates the non-negativity requirement
- Therefore, the guess is invalid

5 Intuitive Explanation

5.1 What Does $\lambda_1 < 0$ Mean?

Remember: λ_1 is the shadow price of the constraint $x_1 \geq 0$.

- $\lambda_1 > 0$ means: "I'd benefit from relaxing $x_1 \geq 0$ (buying negative x_1)"
- $\lambda_1 = 0$ means: "The constraint $x_1 \geq 0$ doesn't limit me"
- $\lambda_1 < 0$ means: "Relaxing $x_1 \geq 0$ would HURT me" - but this is nonsense!

The contradiction: If $\lambda_1 < 0$, it means you'd be WORSE OFF if you could buy $x_1 < 0$ (negative good 1). But you're already at $x_1 = 0$, so this constraint isn't actually binding you in a meaningful way.

5.2 The Economic Intuition

Why the Guess Was Wrong

The guess: Buy only good 2 ($x_1 = 0, x_2 = u^*$)

The reality: When $p_1 < p_2$ (good 1 is CHEAPER), you should buy only good 1, not good 2!

What went wrong: We guessed that the consumer buys the MORE EXPENSIVE good, which is obviously not optimal for cost minimization.

5.3 Concrete Example

Suppose:

- Coffee costs \$2 per cup ($p_1 = 2$)
- Tea costs \$5 per cup ($p_2 = 5$)
- You need utility 10 ($u^* = 10$)

The wrong guess: Buy 0 coffee, 10 teas

- Cost: $2(0) + 5(10) = \$50$

The right choice: Buy 10 coffees, 0 teas

- Cost: $2(10) + 5(0) = \$20$

Obviously, buying only the MORE EXPENSIVE good (\$50) is worse than buying only the CHEAPER good (\$20)!

When we apply the math:

$$\lambda_1 = p_1 - p_2 \tag{11}$$

$$= 2 - 5 \tag{12}$$

$$= -3 < 0 \quad \text{INVALID!} \tag{13}$$

6 What Should the Correct Guess Be?

When $p_1 < p_2$ (good 1 is cheaper), the correct guess should be:

- $x_1 = u^*$ (buy only good 1)
- $x_2 = 0$ (buy no good 2)
- Nonnegativity on good 1 is NON-binding ($x_1 > 0$ so $\lambda_1 = 0$)
- Nonnegativity on good 2 IS binding ($x_2 = 0$ so $\lambda_2 \geq 0$)

This is the OPPOSITE of what we guessed in part (b)!

7 Complete Answer to Problem 2c

Problem 2c Answer

Showing the solution is invalid:

Using the guess from part (b) with $p_1 < p_2$, we get:

$$x_1 = 0 \quad (14)$$

$$x_2 = u^* \quad (15)$$

$$\lambda_1 = p_1 - p_2 < 0 \quad (\text{INVALID} - \text{violates } \lambda_1 \geq 0) \quad (16)$$

$$\lambda_2 = 0 \quad (17)$$

$$\lambda_3 = p_2 \quad (18)$$

Since $\lambda_1 < 0$, this violates the KKT conditions. The solution is invalid.

Intuitive explanation:

The guess assumed we buy only good 2 ($x_1 = 0, x_2 = u^*$). But when $p_1 < p_2$, good 1 is CHEAPER, so buying only the more expensive good 2 is clearly not cost-minimizing.

The negative λ_1 tells us mathematically what we know intuitively: we should be buying MORE of good 1 (the cheaper good), not zero.

The correct guess when $p_1 < p_2$ should have nonnegativity on good 2 binding (buy only good 1), not nonnegativity on good 1 binding.

8 General Pattern for Perfect Substitutes

The Rule

For perfect substitutes:

- If $p_1 < p_2$: Buy only good 1 ($x_1 = u^*, x_2 = 0$)
- If $p_1 > p_2$: Buy only good 2 ($x_1 = 0, x_2 = u^*$)
- If $p_1 = p_2$: Any combination with $x_1 + x_2 = u^*$ works

The KKT conditions will only be satisfied (all $\lambda_i \geq 0$) when you guess the RIGHT corner solution.

9 Summary

Key Takeaway

The point of Problem 2c: Show that the Lagrangian method catches our mistakes!

If you guess the wrong corner solution:

1. The math will produce a negative Lagrange multiplier
2. This violates the KKT conditions
3. This tells you your guess was wrong
4. You need to try a different guess

This is how you systematically check all possible corner solutions until you find the right one.